



PIR-100

LoRaWAN PIR Sensor

User Manual



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1. Description

1.1 Product Description

The PIR-100 is a LoRaWAN-based PIR sensor designed for detecting human infrared radiation. Utilizing advanced infrared detection and analysis technology, it can automatically detect human activity within its range. Upon detecting an intruder passing through the secured area, it immediately sends an alarm signal to the monitoring platform.

It is widely applicable to various settings such as residential areas, villas, factories, shopping malls, warehouses, and office buildings.

1.2 Features

- ✧ Support LoRaWAN Protocol
- ✧ Support wireless remote real time alarm
- ✧ Ceiling-mounted or wall-mounted installation
- ✧ Detecting distance 8-10m
- ✧ Support Anti-Tamper Alarm
- ✧ Support Low Power Alarm
- ✧ Support Status Report
- ✧ Built-in battery with 5 years lifespan

1.3 Specification

Item	Description
Antenna	Built-in LoRa Antenna
Transmission Rate	300bps-5.4Kbps
Frequency Optional	CN470-510MHz EU863-870MHz US902-928MHz AS923MHz AU915-928MHz KR920-923MHz
Maximum Transmission Power	19dBm
Rx Sensitivity	-141dBm(SF=12)
Detecting Range	8-10m
Detecting Angle	50°~90°
Installation Height	1.5m~2.0m
Working Temperature	-10℃- +55℃
Working Humidity	10%-95% (no condensing)
Power Supply Method	Built-in Li-SOCL2
Battery Capacity	8000mAh
Supply Voltage	DC 3.6V
Sleep Current	≤ 10uA
Detection Current	≤ 25uA
Transmission current	≤ 125mA
Dimension	86*86*26mm

2. Structure

2.1 Appearance and dimension



Figure: PIR Sensor

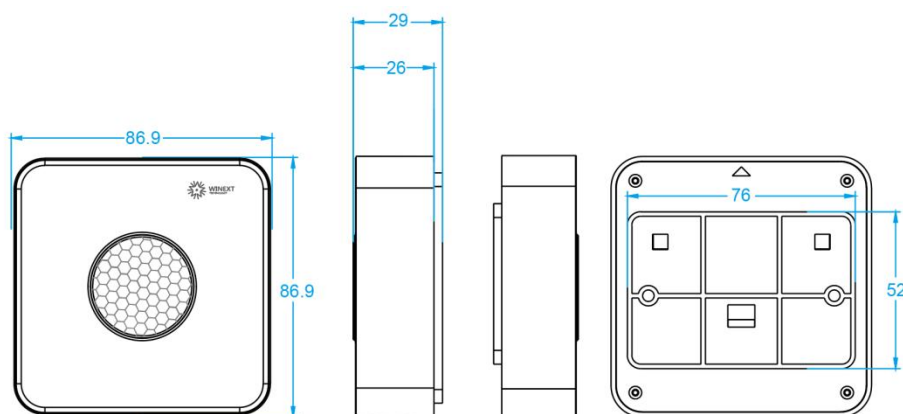


Figure: Dimension(mm)

2.2 Packing List

Accessory	Qty(pc)	Remark
PIR Motion Sensor	1	/
Installation board	1	/
Screw	4	2pcs long plastic plug & 2pcs short screw
3M double-side tape	1	/
Conformity Certificate	1	/

3. Operation

- Power On: Press and hold the power button until the green light turns on and then off. Release the button after it goes out.
- Power Off: Press and hold the button for more than 3 seconds until the red light stays on. Release the button after the red light turns off.
- Joining Network: After powering on, if the green light flashes 3 times consecutively, it indicates the device has successfully joined the network.
- Power-on Status Check: While the device is on, briefly press the power button once; the red light will flash once.
- Infrared Detection: When human body is detected, the green light will flash once.
- Data Reporting: When LoRaWAN data is sent or received, the red light will flash once.

Note: After the infrared alarm is powered on, the detector enters normal working mode. It immediately sends an "occupancy detected" alarm event upon detecting human movement. After sending this alert, it will not report further movement if the person remains within the monitoring range. If no human presence is detected for 2 minutes, it will report an "unoccupied" data packet. By default, the device sends a heartbeat data packet every 8 hours.

4. Installation

4.1 Safety precaution

To prevent equipment damage or false alarms caused by improper installation, please adhere to the following precautions:

- ✧ Ensure the device is powered ON before installation.
- ✧ Installation must be performed by professional installers or authorized engineering representatives.
- ✧ The recommended installation height for the infrared alarm is 1.5 - 2.1 meters.
- ✧ Do not point the device directly at HVAC vents or sources of hot/cold air (e.g., high-power appliances, electric heaters).
- ✧ Do not aim the device directly at glass doors or windows to avoid interference from strong light.

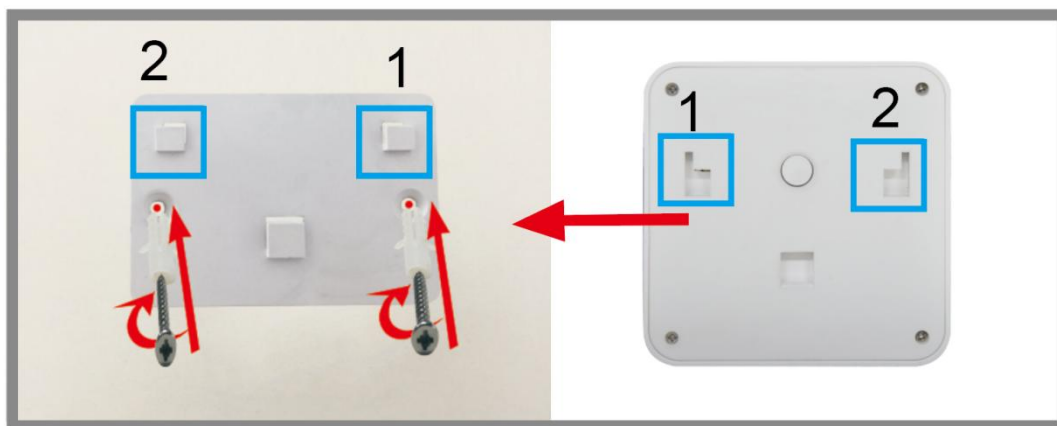
4.2 Tools

Before installation, please prepare proper tools. Here are tools may be used for reference:



Figure: Installation Tools

4.3 Installation Instructions

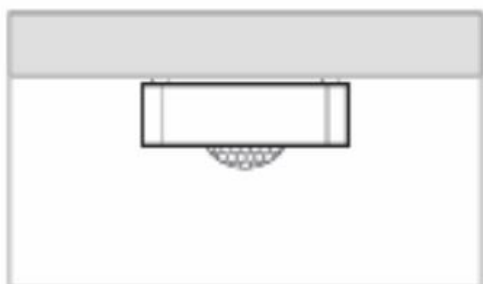


Place the back board on the position and mark it where to be installed. Use the percussion drill to make punch holes. Put the plastic cover into the 2 holes and use the tapping screws to secure the back on the wall.

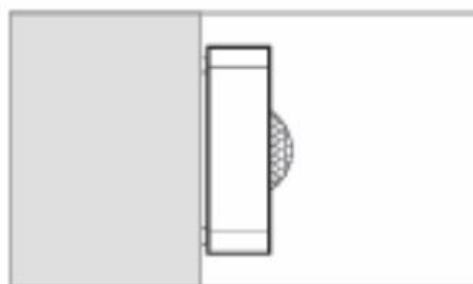
Note: When installing on metal objects, glass walls, or surfaces where drilling holes is not possible, you can use 3M double-sided adhesive tape for mounting. Please follow the steps below:

- Use a dry towel to clean up on the surface to install.
- Make the 3M tape adhere to the device back board. Then make the device back board and the device back holder fit together well.
- Locate PIR-100 on a proper position where 3M tape to be attached. Press several times to make sure it gets fixed. Recommended installation height is at least 1.5 m.

4.4 Installation Method



Ceiling-mounted



Wall-mounted

5. Communication Format

Fort Port is 210. Data is displayed in hexadecimal format, with a length of 12 bytes.

Uplink Data Command Basic Format:

Channel 1	Channel 2	Product Model	----	Channel 5	Alarm Event
1 Byte	1 Byte	1 Byte	----	1 Byte	1 Byte

Heartbeat data reporting interval: 8 hours

Example using Infrared Alarm Data: 000104040c857d0077011701

5.1 Heartbeat frame

Channel Number	Type (Byte Length)	Example Data	Parsing Explanation
00	Reserved (1 Byte)	00	00 is a reserved channel
01	Product Model (1 Byte)	04	04 corresponds to sensor
04	Battery Voltage (2 Bytes)	0c85	0c85 converted to decimal is 3205. Divide by 1000 to get 3.205V
7d	Voltage Status (1 Byte)	00	00: Normal; 01: Low Voltage
77	Tamper Status (1 Byte)	01	00: Normal; 01: Tampered
05	Low Battery Alarm (1 Byte)	00	00: Normal; 01: Alarm
03	Tamper Alarm (1 Byte)	01	00: Normal; 01: Tamper Alarm
17	Infrared Alarm (1 Byte)	01	00: Normal; 01: Alarm

An aerial night view of a city with illuminated skyscrapers and a complex highway interchange. Overlaid on the image is the LoRaWAN logo, which consists of the text "LoRaWAN" in a stylized font with a signal wave icon to the left. Several concentric blue circles and arcs represent signal waves emanating from the logo and across the city scene.

LoRaWAN™

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